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Thermodynamic Assessments of the Bi-Tb and Bi-Y Systems

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By using the calculation of phase diagrams (CALPHAD) method, the thermodynamic assessments of the Bi-Tb and Bi-Y systems were carried out based on the available experimental data including thermodynamic properties and phase equilibria. Gibbs free energies of the liquid, hcp, bcc, and rhombohedral phases in the Bi-Tb and Bi-Y systems were modeled by the substitutional solution model, and the intermetallic compounds (BiTb, Bi_3Tb_4, αBi_3Tb_5, βBi_3Tb_5, BiY, and Bi_3Y_5 phases) in these two binary systems were described by the sublattice model. An agreement between the present calculated results and experimental data was obtained.

Keywords CALPHAD, phase diagram, thermodynamic assessment, thermodynamic properties

1. Introduction

Rare earth (RE) elements are added into the Mn-Bi system to form the RE_3MnBi_5 and $RE_2Mn_3Bi_6$ ternary compounds with good electrical and magnetic properties.[1] Moreover, RE elements are also important for Bi-based superconductors which have great technological interest.[2] The binary Bi-RE systems are essential systems for the above-mentioned materials. Therefore, it is important to understand the phase equilibria in the Bi-RE binary system.

The calculation of phase diagrams (CALPHAD) method, which is a powerful tool to cut down on cost and time during development of materials,[3] effectively provides a clear guideline for materials design. In order to design high-performance alloys with RE elements, it is important to develop a thermodynamic database including RE elements alloys. Currently, our group is focusing on the development of a thermodynamic database of RE alloy system,[4-13] which are important for the design of alloys with additions of RE elements. As a part of this thermodynamic database, this present work is to present the thermodynamic descriptions of the Bi-Tb and Bi-Y binary systems based on the available experimental data by means of the CALPHAD method.

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2. Experimental Information

2.1 The Bi-Tb System

The Bi-Tb system consists of three terminal solid solution phases (αTb, βTb, and rhombohedral (Bi) phases) and four intermetallic compounds (BiTb, Bi_3Tb_4, αBi_3Tb_5, and βBi_3Tb_5 phases). Based on differential thermal analysis, x-ray diffraction and microstructural analysis, Abulkhaev[14] investigated the phase diagram of the Bi-Tb system, and reported three stoichiometric compounds: BiTb (melting point, 1760 ± 15 °C), Bi_3Tb_4 (peritectic formation at 1600 ± 15 °C) and Bi_3Tb_5 (peritectic formation at 1490 ± 15 °C). Abulkhaev[14] also determined a metatectic reaction (βTb (99 at.% Tb) ↔ L + αTb (> 99 at.% Tb) at 1538 ± 10 °C) and two eutectic reactions: L (0.2 at.% Tb) ↔ (Bi) + BiTb at 269 ± 2 °C and L (83 ± 1 at.% Tb) ↔ αTb (> 99 at.% Tb) + αBi_3Tb_5 at 1105 ± 10 °C. A reversible polymorphic transition ($\alpha Bi_3Tb_5 \leftrightarrow \beta Bi_3Tb_5$ at 1410 ± 10 °C) was observed by Abulkhaev.[14] In the work of Abulkhaev,[14] the solid solubility of Bi in the αTb phase was suggested to be less than 1.0 at.%, and the solid solubility of Tb in (Bi) phase was indicated to be negligible. The phase diagram of the Bi-Tb system reviewed by Okamoto[15] is shown in Fig. 1.

In addition, Colinet et al.[16] predicted the standard enthalpies of formation of the BiTb, Bi_3Tb_4, and Bi_3Tb_5 phases by using Miedema's model. With the same method, Colinet et al.[16] and de Boer and Perrifor[17] predicted that the value of the enthalpy of mixing of the liquid phase with the reference states of pure liquid metals at 50 at.% Tb in the Bi-Tb system is respectively −47 kJ/mol[16] and −53 kJ/mol;[17] however, no temperature dependence was given in the work of Colinet et al.[16] and de Boer and Perrifor.[17]

2.2 The Bi-Y System

The Bi-Y system consists of three terminal solid solution phases (αY, βY, and rhombohedral (Bi) phases) and two intermetallic compounds (BiY and Bi_3Y_5 phases). Based on microscopic, x-ray, thermal, and chemical analyses, Schmidt et al.[18] investigated the phase diagram in the Bi-Y system,

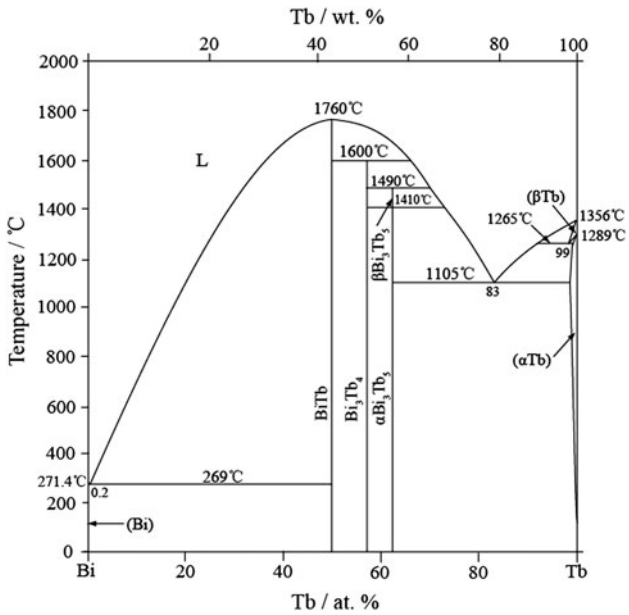

Fig. 1 The phase diagram of the Bi-Tb system reviewed by Okamoto[15]

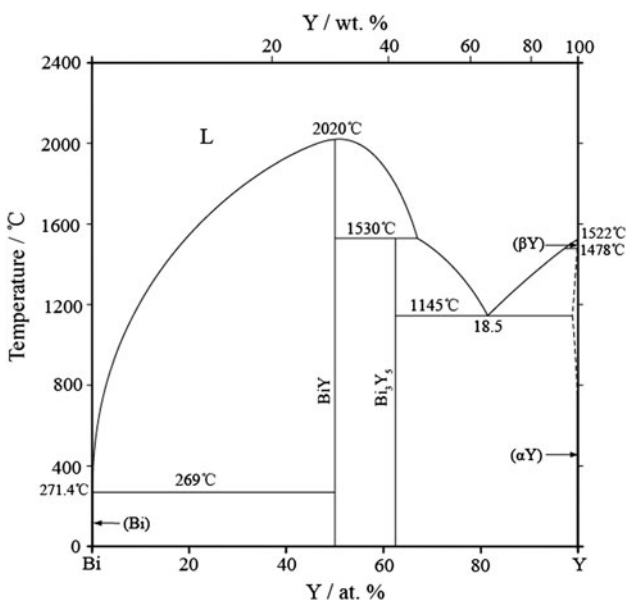

Fig. 2 The phase diagram of the Bi-Y system reviewed by Gschneidner and Calderwood[19]

and reported two stoichiometric compounds: BiY (melting point, 2020 ± 10 °C) and Bi_3Y_5 (peritectic formation at 1530 ± 5 °C). Schmidt et al.[18] also determined two eutectic reactions: L (<0.1 at.% Y) ↔ (Bi) + BiY at 269 °C and L (81.5 at.% Y) ↔ αY + Bi_3Y_5 at 1145 °C. In the work of Schmidt et al.,[18] the solubility of Bi in Y, and Y in Bi were suggested to be negligible. The phase diagram of the Bi-Y system reviewed by Gschneidner and Calderwood[19] is shown in Fig. 2.

In addition, Ferro et al.[20] reported the standard enthalpies of formation of the BiY and Bi_3Y_5 phases by means of a direct isoperibol calorimeter. Using the electromotive force (EMF), Kober et al.[21] measured the enthalpy and entropy of formation of the BiY phase between 387 and 677 °C with different reference states. By using Miedema's model, Colinet et al.[16] and de Boer and Perrifor[17] predicted that the value of the enthalpy of mixing of the liquid phase with the reference states of pure liquid metals at 50 at.% Y in the Bi-Y system is respectively −48 kJ/mol[16] and −54 kJ/mol;[17] however, no temperature dependence was given in the work of Colinet et al.[16] and de Boer and Perrifor.[17]

Table 1 The stable solid phases and the models used in the Bi-Tb[15] and Bi-Y[19] systems

System	Phase	Prototype	Modeling phase	Used models
Bi-Tb	(Bi)	αAs	$(Bi, Tb)_1$	SSM
	BiTb	NaCl	(Bi)(Tb)	SM
	Bi_3Tb_4	Th_3P_4	$(Bi)_3(Tb)_4$	SM
	αBi_3Tb_5	…	$(Bi)_3(Tb)_5$	SM
	βBi_3Tb_5	Mn_5Si_3	$(Bi)_3(Tb)_5$	SM
	(βTb)	W	$(Bi, Tb)_1:(Va)_3$	SSM
	(αTb)	Mg	$(Bi, Tb)_1:(Va)_{0.5}$	SSM
Bi-Y	(Bi)	αAs	$(Bi, Y)_1$	SSM
	BiY	NaCl	(Bi)(Y)	SM
	Bi_3Y_5	…	$(Bi)_3(Y)_5$	SM
	(βY)	W	$(Bi, Y)_1:(Va)_3$	SSM
	(αY)	Mg	$(Bi, Y)_1:(Va)_{0.5}$	SSM

SSM: substitutional solution model; SM: sublattice model

3. Thermodynamic Models

Information about stable solid phases and the models used in the Bi-Tb and Bi-Y binary systems is listed in Table 1.

3.1 Solution Phases

In the Bi-Me (Me: Tb or Y) binary system, the Gibbs free energies of the liquid, rhombohedral, bcc, and hcp phases are described by the substitutional solution model, as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{Bi,Me}} G_i^{\phi} x_i + RT \sum_{i=\mathrm{Bi,Me}} x_i \ln x_i + {}^{\mathrm{E}}G_m^{\phi}, \qquad \text{(Eq 1)}$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component i in the respective reference state with ϕ phase, which is taken from the SGTE pure element database;[22] x_i denotes the mole fraction of component i; R is the gas constant; T is the absolute temperature; and the term ${}^0G_m^{\phi}$ presents the excess Gibbs free energy, which is expressed in Redlich-Kister polynomial form[23] as:

$${}^{\mathrm{E}}G_m^{\phi} = x_{\mathrm{Bi}} x_{\mathrm{Me}} \sum_{m=0}^{n} {}^m L_{\mathrm{Bi,Me}}^{\phi} (x_{\mathrm{Bi}} - x_{\mathrm{Me}})^m, \qquad \text{(Eq 2)}$$

$$^{m}L^{\phi}_{\mathrm{Bi,Me}} = a + bT, \quad \text{(Eq 3)}$$

where $^{m}L^{\phi}_{\mathrm{Bi,Me}}$ is the binary interaction parameter, and the coefficients of a and b are evaluated on the basis of available experimental data.

3.2 Stoichiometric Intermetallic Compounds

The intermetallic compounds of the BiTb, Bi_3Tb_4, αBi_3Tb_5, βBi_3Tb_5, BiY, and Bi_3Y_5 phases in the Bi-Me (Me: Tb or Y) binary system are all treated as stoichiometric phases. The Gibbs free energy per mole of formula unit Bi_pMe_q can be expressed as follows:

$$\Delta^0 G_f^{Bi_pMe_q} = G_m^{Bi_pMe_q} - p^0 G_{Bi}^{SER} - q^0 G_{Me}^{SER} = a' + b'T, \quad \text{(Eq 4)}$$

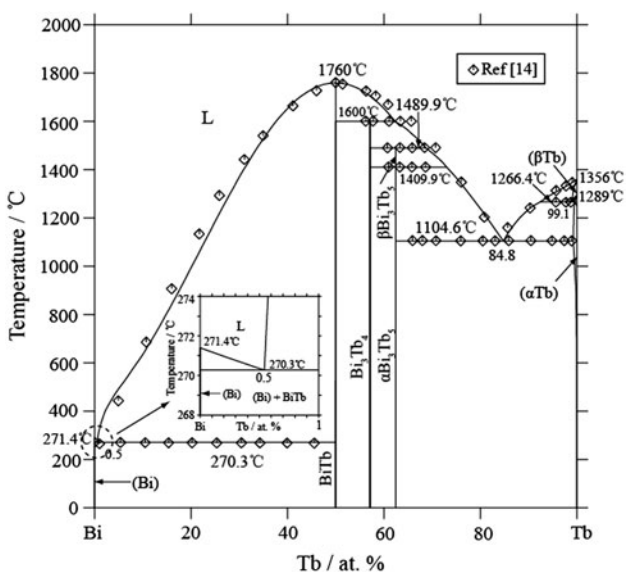

Fig. 3 Calculated phase diagram of the Bi-Tb system compared with the experimental data[14]

where $\Delta^0 G_f^{Bi_pMe_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements; the parameters of a' and b' are evaluated in the present work.

4. Optimized Results and Discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software,[24] which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of the data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

4.1 The Bi-Tb System

The calculated Bi-Tb phase diagram compared with the experimental data is shown in Fig. 3, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 2. It is seen that the calculated results are in reasonable agreement with the experimental data.[14] A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Bi-Tb system is given in Table 3.

The calculated standard enthalpies of formation with reference states of the Bi (rhombohedral) and αTb (hcp) phases compared with the predicted data[16] are presented in Fig. 4. It is seen that a reasonable agreement for the thermodynamic properties is obtained between the calculated results and the predicted data.[16] Figure 5 shows the calculated enthalpies of mixing of the liquid phase at 2000 °C with reference states of pure Bi (liquid) and Tb (liquid) in the Bi-Tb binary system. The calculated value at

Table 2 Experimental and calculated invariant reactions in the Bi-Tb system

Invariant reaction	Reaction type	Composition, at.% Tb			Temperature, °C	Reference
L ↔ (Bi) + BiTb	Eutectic	0.2	0	50	269 ± 2	[14]
		0.5	0	50	270.3	This work
L ↔ BiTb	Congruent		50.0		1760 ± 15	[14]
			50.0		1760	This work
L + BiTb ↔ Bi_3Tb_4	Peritectic		50.0	57.1	1600 ± 15	[14]
		62.3	50.0	57.1	1600	This work
L + Bi_3Tb_4 ↔ βBi_3Tb_5	Peritectic		50.0	57.1	1490 ± 15	[14]
		68.7	50.0	57.1	1489.9	This work
αBi_3Tb_5 ↔ βBi_3Tb_5	Polymorphic		62.5		1410 ± 10	[14]
			62.5		1409.9	This work
L ↔ αBi_3Tb_5 + αTb	Eutectic	83 ± 1	62.5	> 99	1105 ± 10	[14]
		84.8	62.5	99.2	1104.6	This work
βTb ↔ L + αTb	Metatectic	> 99		> 99	1265 ± 10	[14]
		99.1	92.4	99.5	1266.4	This work

Table 3 Thermodynamic parameters in the Bi-Tb system assessed in the present work

Parameters in each phase, J/mol
Liquid phase, format (Bi, Tb)
${}^{0}L^{\mathrm{Liq}}_{\mathrm{Bi,Tb}} = -198749 + 24.883T$
${}^{1}L^{\mathrm{Liq}}_{\mathrm{Bi,Tb}} = 2000 - 1.6T$
${}^{2}L^{\mathrm{Liq}}_{\mathrm{Bi,Tb}} = 39500 + 10T$
Bcc phase, format (Bi, Tb)
${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Bi,Tb}} = -101000$
Hcp phase, format (Bi, Tb)
${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Bi,Tb}} = -94000$
Rhombohedral phase, format (Bi, Tb)
${}^{0}G^{\mathrm{Rhombohedral}}_{\mathrm{Tb}} = 5000 + {}^{0}G^{\mathrm{Hcp}}_{\mathrm{Tb}}$
${}^{0}L^{\mathrm{Rhombohedral}}_{\mathrm{Bi,Tb}} = 0$
BiTb phase, format $(Bi)_{0.5}(Tb)_{0.5}$
$\Delta^{0}G^{\mathrm{BiTb}}_{\mathrm{f}} = -82080 + 7.874T$
Bi_3Tb_4 phase, format $(Bi)_{0.429}(Tb)_{0.571}$
$\Delta^{0}G^{\mathrm{Bi_3Tb_4}}_{\mathrm{f}} = -74520 + 7.012T$
αBi_3Tb_5 phase, format $(Bi)_{0.375}(Tb)_{0.625}$
$\Delta^{0}G^{\alpha\mathrm{Bi_3Tb_5}}_{\mathrm{f}} = -64805 + 4.873T$
βBi_3Tb_5 phase, format $(Bi)_{0.375}(Tb)_{0.625}$
$\Delta^{0}G^{\beta\mathrm{Bi_3Tb_5}}_{\mathrm{f}} = -63725 + 4.231T$

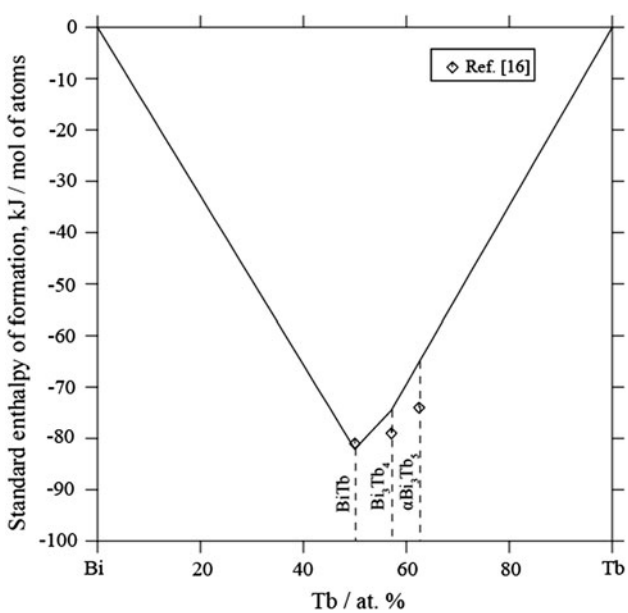

Fig. 4 Calculated standard enthalpies of formation with reference states of Bi (rhombohedral) and αTb (hcp) phases in the Bi-Tb system compared with the predicted data[16]

50 at.% Tb is −49.8 kJ/mol, which is close to the predicted values of −47 kJ/mol[16] and −53 kJ/mol.[17] Therefore, the calculated enthalpies of mixing of the liquid phase in the Bi-Tb system are reasonable.

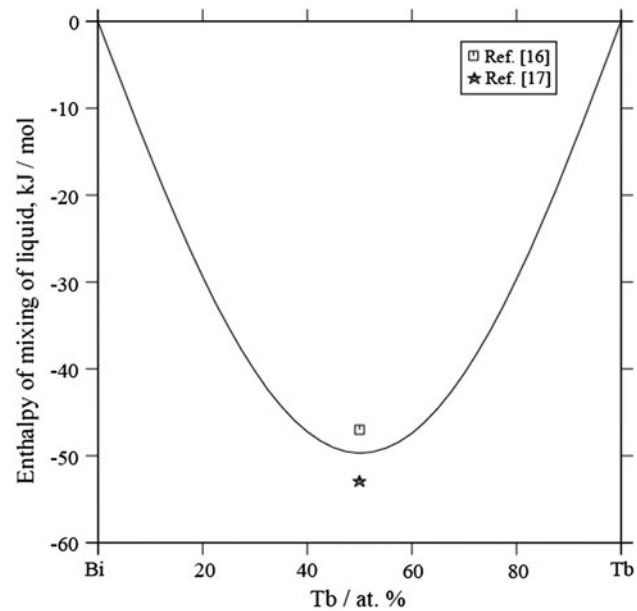

Fig. 5 Calculated enthalpies of mixing of liquid phase at 2000 °C with reference states of pure Bi (liquid) and Tb (liquid) compared with the predicted data[16,17]

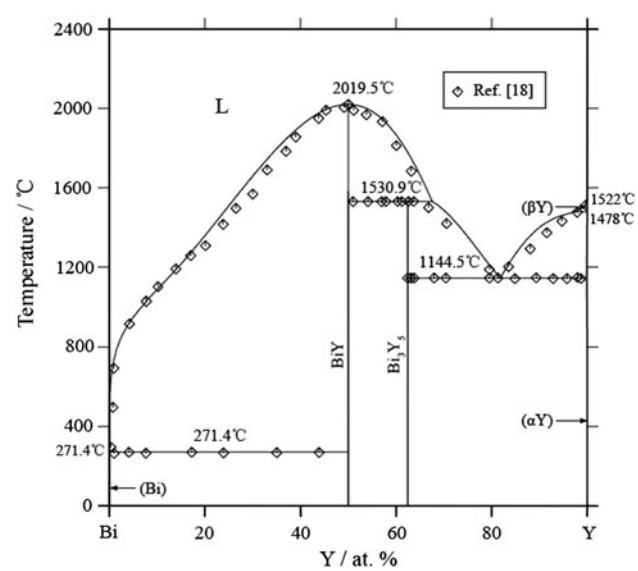

Fig. 6 Calculated phase diagram of the Bi-Y system compared with the experimental data[18]

4.2 The Bi-Y System

The calculated Bi-Y phase diagram compared with the experimental data is shown in Fig. 6, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 4. It is seen that the calculated results are in

Table 4 Experimental and calculated invariant reactions in the Bi-Y system

Invariant reaction	Reaction type	Composition, at.% Y			Temperature, °C	Reference
L ↔ (Bi) + BiY	Eutectic	<0.1	0	50	269	[18]
		~0	0	50	271.4	This work
L ↔ BiY	Congruent		50.0		2020 ± 10	[18]
			50.0		2019.5	This work
L + BiY ↔ Bi_3Y_5	Peritectic		50.0	62.5	1530 ± 5	[18]
		67.6	50.0	62.5	1530.9	This work
L ↔ Bi_3Y_5 + (αY)	Eutectic	81.5	62.5	~100	1145	[18]
		81.8	62.5	99.8	1144.5	This work

Table 5 Thermodynamic parameters in the Bi-Y system assessed in the present work

Parameters in each phase, J/mol
Liquid phase, format (Bi, Y)
${}^0L^{Liq}_{Bi,Y} = -209402 - 7.363T$
${}^1L^{Liq}_{Bi,Y} = -23300 + 48.1T$
${}^2L^{Liq}_{Bi,Y} = 143800 - 45T$
${}^3L^{Liq}_{Bi,Y} = -44800$
Hcp phase, format (Bi, Y)
${}^0L^{Hcp}_{Bi,Y} = -123000$
Rhombohedral phase, format (Bi, Y)
${}^0G^{Rhombohedral}_{Y} = 5000 + {}^0G^{Hcp}_{Y}$
${}^0L^{Rhombohedral}_{Bi,Y} = 0$
BiY phase, format $(Bi)_{0.5}(Y)_{0.5}$
$\Delta^0G^{Bi,Y}_f = -90761 + 0.16T$
Bi_3Y_5 phase, format $(Bi)_{0.375}(Y)_{0.625}$
$\Delta^0G^{Bi_3Y_5}_f = -76615 + 0.53T$

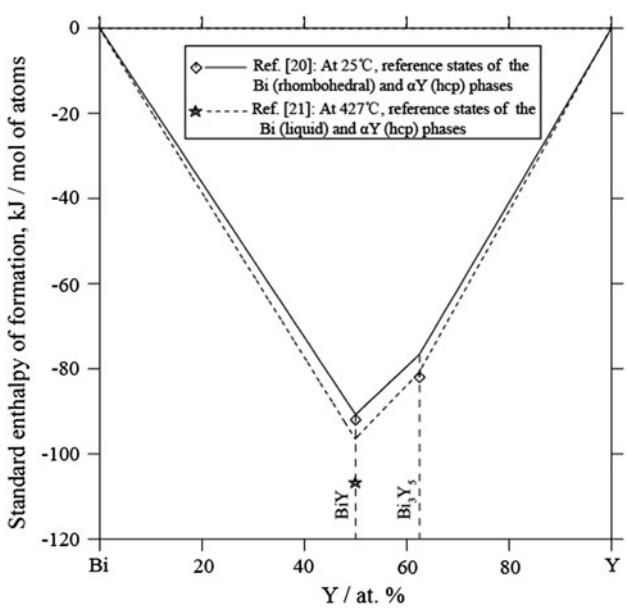

Fig. 7 Calculated enthalpies of formation of intermetallic phases at 25 and 427 °C compared with the experimental data[20,21] in the Bi-Y system

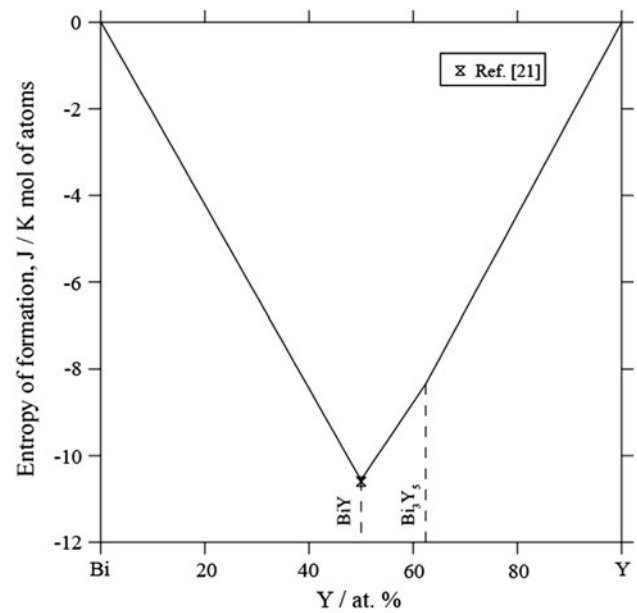

Fig. 8 Calculated entropies of formation of intermetallic phases at 427 °C with reference states of Bi (liquid) and αY (hcp) phases compared with the experimental data[21] in the Bi-Y system

reasonable agreement with the experimental data.[18] A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Bi-Y system is given in Table 5.

The calculated enthalpies and entropies of formation of intermetallic phases compared with the experimental data[20,21] in the Bi-Y system are presented in Fig. 7 and 8. It is seen that an agreement for the thermodynamic properties is obtained between the calculated results and the experimental data, except that the calculated enthalpy of formation of the BiY phase at 427 °C is less negative than the value obtained by Kober et al.[21] By considering the agreements of the most of the experimental data including the data of phase equilibria and thermodynamic properties, we think that this calculated result can be accepted. Figure 9 shows the calculated enthalpies of mixing of the liquid phase at 2200 °C with reference states of pure Bi (liquid) and Y (liquid) in the Bi-Y binary system. The calculated value at 50 at.% Y is −52.35 kJ/mol, which is close to the predicted values of −48 kJ/mol[16] and −54 kJ/mol.[17]

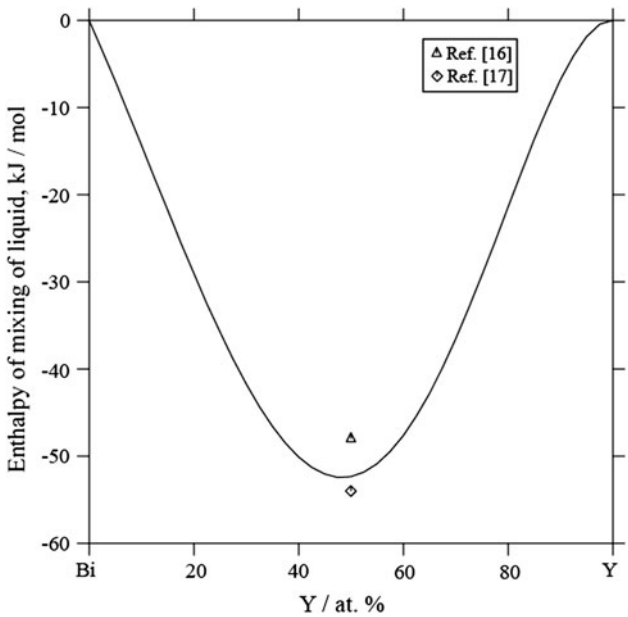

Fig. 9 Calculated enthalpies of mixing of liquid phase at 2200 °C with reference states of pure Bi (liquid) and Y (liquid) compared with the predicted data[16,17]

Therefore, the calculated enthalpies of mixing of the liquid phase in the Bi-Y system are reasonable.

5. Conclusions

The phase diagrams in the Bi-Tb and Bi-Y binary systems were thermodynamically assessed based on experimental data including phase equilibria and thermodynamic data, and most of the experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters.

Acknowledgments

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